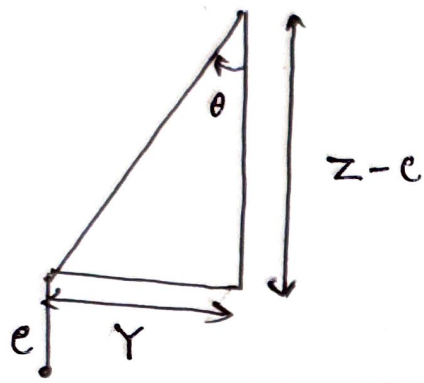
I.K

$x = x_f - x_h$, $y = y_f - y_h$, $z = z_f - z_h$

X – Forward
Y – Lateral
Z – vertical

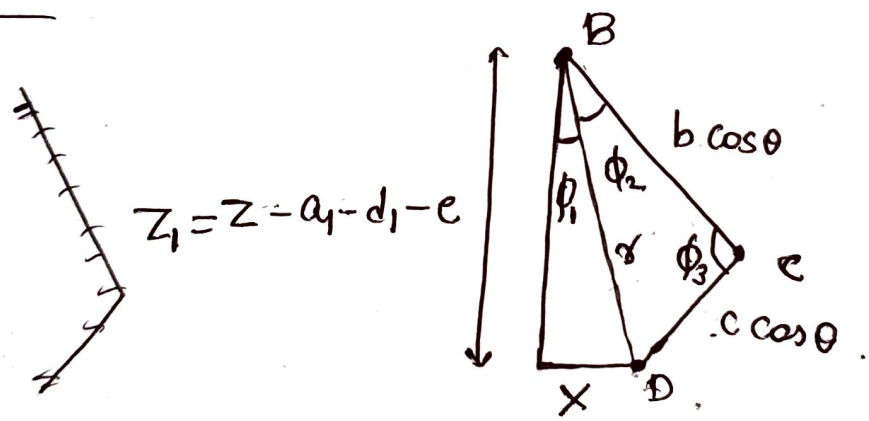
Front view.



Note, we are ammending the angle iks in a 5X1 matrix $q_5 = [0, 0, 0, 0, 0]$

$$\boxed{q_0 = -\tan^{-1}\left(\frac{Y}{Z-e}\right)}, \quad \boxed{q_4 = q_0}$$

Side view



$Z_1 = Z - a_1 - d_1 - e$

$$\boxed{\phi_1 = \tan^{-1}\left(\frac{X}{Z_1}\right)}$$

$$r = \sqrt{Z_1^2 + X^2}$$

$$b_1 = b\cos\theta.$$

$$c_1 = c\cos\theta.$$

$$\boxed{\phi_2 = \cos^{-1}\left(\frac{b_1^2 + r^2 - c_1^2}{2 b_1 r}\right)}$$

$$q_3 = \phi_1 + \phi_2$$

$$\boxed{\phi_3 = \cos^{-1}\left(\frac{b_1^2 + c_1^2 - r^2}{2 b_1 c_1}\right)}$$

$$q_2 = \pi - \phi_3$$

$$q_1 = -q_3 + q_2$$